A4.2 Conservation of biodiversity

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-a4-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 156–180 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Biodiversity, past and present

A4.2.1 Biodiversity in all its forms and levels

pp. 156–159

Biodiversity includes **ecosystem** diversity, **species** diversity and **genetic** diversity. Species diversity depends on **richness**, the number of species, and **evenness**, how similar their numbers are. **Simpson's reciprocal index** compares communities: $D = N(N - 1) / \Sigma n(n - 1)$, and a higher value means higher diversity.

A4.2.2 Today compared with the past

pp. 160–161

Millions of species have been named, and many more remain to be discovered. Fossils suggest that more species are alive now than at any time in the past, although past levels are hard to estimate.

What is being lost, and why

A4.2.3 Human-caused extinction

pp. 162–164

Species	Causes
Moa, New Zealand	Hunted, and their forest burnt and cleared, after people arrived around 1280; giant moa were gone by about 1500
Caribbean monk seal	Hunted for oil and meat, and last confirmed in 1952

Add a third brief case study from your own region.

A4.2.4 Causes of ecosystem loss

pp. 164–167

Causes include farming, logging, mining, dams, draining water, pollution and climate change. **Mixed dipterocarp forest** in South East Asia is being lost to timber and plantations such as oil palm. Your local example could be Swiss moorland or river floodplains.

A4.2.5 Evidence of a biodiversity crisis

p. 168

Reports from IPBES and other sources draw on surveys of many habitats and groups. They include population sizes, species ranges, richness and evenness, and how degraded ecosystems are. Citizen scientists can add data too.

A4.2.6 Causes of the current crisis

pp. 169–173

Human population growth is the overarching cause. Specific causes are hunting and over-exploitation, urbanization, deforestation and land clearance, pollution, and invasive species.

What can be done

A4.2.7 Why several approaches are needed

pp. 174–178



Why several approaches are needed

A4.2.7

	In situ	Ex situ
Meaning	Within natural habitats	Outside natural habitats
Examples	Reserves, rewilding	Zoos, seed banks
Strength	Protects whole ecosystems	Saves critically rare species
Weakness	Needs land and enforcement	Small gene pools; behaviour lost

Key idea
Habitat protection is usually most important; ex situ is a safety net.

A4.2 Conservation of biodiversity – Textbook pp. 174–176 20

No single approach is enough. **In situ** methods include nature reserves, rewilding and reclaiming degraded land. **Ex situ** methods include zoos, botanic gardens and seed banks.

A4.2.8 The EDGE of Existence programme

pp. 178–179

EDGE gives priority to species that are **Evolutionarily Distinct** and **Globally Endangered**. With limited funds, saving a species with few close relatives protects more unique evolutionary history.

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